

Sampling Distributions, examples

CSU, Chico Math 314

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outline

Central Limit Theorem

Estimate Bias

References

Representing the CLT

Assume $X_1, \dots, X_n \sim_{iid} G(\theta)$. Simulate the sampling distribution of the point estimator: $T(x)$.

1. Randomly sample n observations from $G(\theta)$ – choose appropriate parameters.
2. Calculate and store $T_r(\mathbf{X})$.
3. Repeat steps 1 and 2 R times.
4. Make a histogram of $T_1, \dots, T_R$.

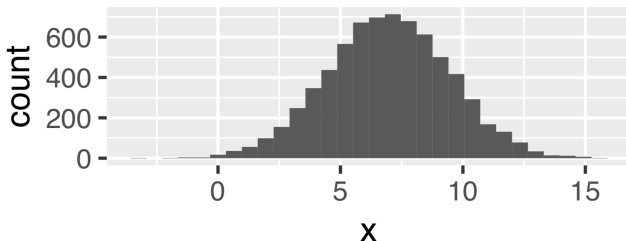
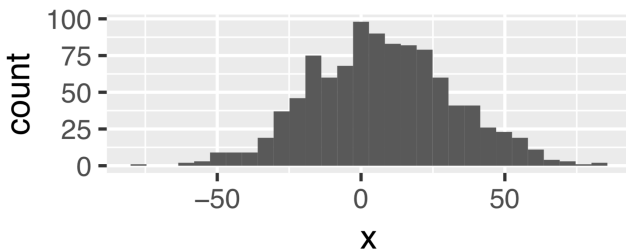
Normal Example (Comparing X and $\bar{X}$'s distributions)

```
R <- 7000
out <- rep(NA, R)
for(i in 1:R){
  ## assume population
  X <- rnorm(100, mean=7, sd=25)
  out[i] <- mean(X) # samples from sampling distrib.
}
X <- rnorm(1e6, 7, 25) # "population" to compare
pop_plot <- ggplot(data.frame(x=X), aes(x=x)) +
  geom_histogram(bins=30)
sampdist_plot <- ggplot(data.frame(x=out), aes(x=x)) +
  geom_histogram(bins=30)
```

NOTE: "population" here isn't *exactly* the starting distribution's PDF—it's just large enough to be a reasonable representation.

Normal Example (Two Plots in One)

```
library(gridExtra)  
grid.arrange(pop_plot, sampdist_plot, nrow=2)
```

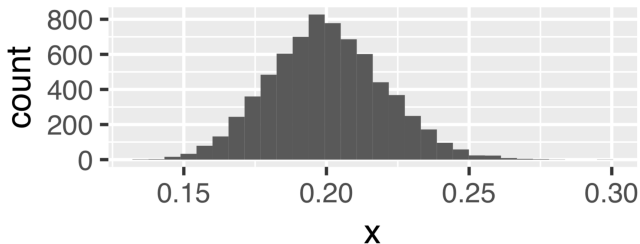
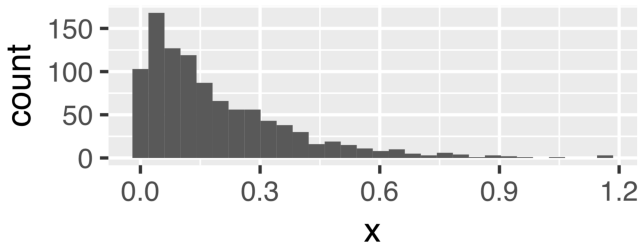


Exponential Example

```
R <- 7000
out <- rep(NA, R)
for(i in 1:R){
  ## assume population
  X <- rexp(100, 5)
  out[i] <- mean(X) # samples from sampling distrib.
}
X <- rexp(1e6, 5) # "population"
pop_plot <- ggplot(data.frame(x=X), aes(x=x)) +
  geom_histogram(bins=30)
sampdist_plot <- ggplot(data.frame(x=out), aes(x=x)) +
  geom_histogram(bins=30)
```

Exponential Example

```
grid.arrange(pop_plot, sampdist_plot, nrow=2)
```



Bias

Recall, bias is defined as $Bias(\hat{\theta}) = E(\hat{\theta}) - \theta$. How does one estimate $E(\hat{\theta})$?

Estimate Bias

Assume $X_1, \dots, X_n \sim_{iid} \text{Bernoulli}(p)$. Estimate the bias of the estimator $\bar{X}$ – as calculated by `optim`. Steps:

1. Randomly sample n observations from $\text{Bernoulli}(p)$: `z <- rbinom(n, 1, p)`.
2. Estimate p .
3. Calculate and store bias: $\text{bias}_r = \hat{p} - p$
4. Repeat steps 1 through 3 R times and calculate $B = R^{-1} \sum_{r=1}^R \text{bias}_r$.

Bernoulli Example

```
pop_p <- 0.654
R <- 7000
out <- rep(NA, R)
for(i in 1:R){
  ## assume population
  X <- rbinom(100, 1, pop_p)
  # Recall: Bias =  $E(\bar{X}) - \text{population parameter}$ 
  out[i] <- mean(X) - pop_p # Point estimates of Bias
}
mean(out) # mean here should be near zero

## [1] -0.001315714
```

references I

Adapted from notes by Dr. Roualdes.